Algebra and Discrete Mathematics
ADM_A_B

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Course Outline

- Vectors and matrices
- System of linear equations
- Matrix inverse and determinants
- Vector spaces, bases and matrix transformations
- Quiz one
- Matrix transformations, fundamental spaces and decompositions
- Eulerian tours and Hamiltonian cycles
- Shortest Paths, Project Scheduling, and Networks
- Quiz two
- Network Flows, Trees, and Matchings
- Make-up Quizzes
- Tutorial 12

Recommended reading

- Saoub, K. R. (2017). A tour through graph theory. Chapman and Hall/CRC.
  - Sections 5.4, 5.5
  - Accessible online (free copy)
  - Alternative download link

Lecture outline

- Matching and vertex cover
- Distinct representatives
- Matchings in Non-Bipartite
- Gale-Shapley Algorithm

Tutorial 12

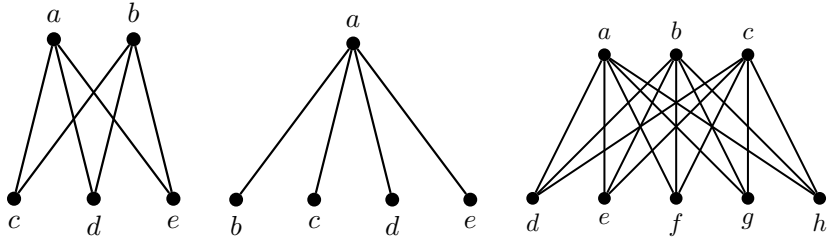
- Matching and vertex cover
- Distinct representatives
- Matchings in Non-Bipartite
- Gale-Shapley Algorithm

Question 1

Question

Draw the complete bipartite graphs $K_{2,3}$, $K_{1,4}$, and $K_{3,5}$.

Solution.



Question 2

Question

- Three student organizations (Student Government, Math Club, and the Equestrian Club) are holding meetings on Thursday afternoon.
- The only available rooms are 105, 201, 271, and 372.
- Based on membership and room size, the Student Government can only use 201 or 372, Equestrian Club can use 105 or 372, and Math Club can use any of the four rooms.
- Find a maximum matching for this scenario.

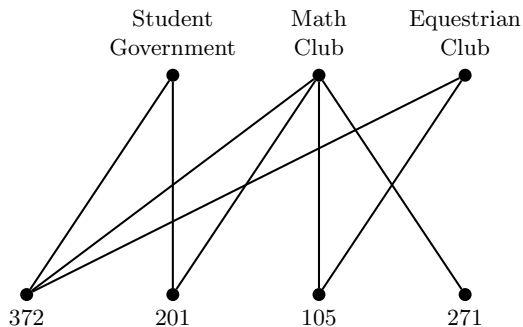
Solution. Each organization and room is represented by a vertex, and an edge denotes when an organization is able to use a room

Question 2

Question

- Student Government can only use 201 or 372, Equestrian Club can use 105 or 372, and Math Club can use any of the four rooms.

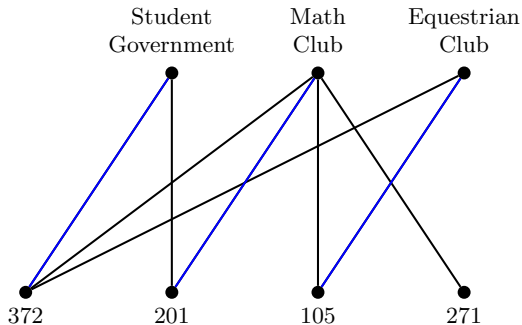
Solution. Each organization and room is represented by a vertex, and an edge denotes when an organization is able to use a room



Question 2

Solution.

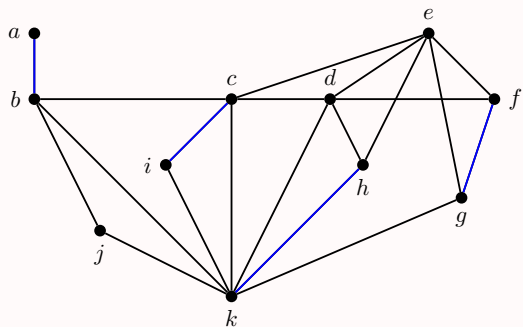
- A matching has at most 3 edges
- Maximum matching



Question 3

Question

Below is a graph with a matching M shown in blue.



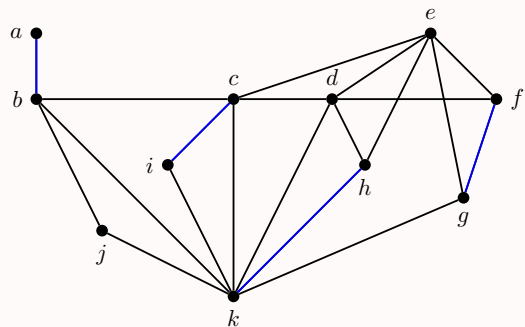
1. Find an alternating path starting at a. Is this path augmenting?

Solution.

1. One alternating path starting at a is $abcikhe$. This path is not augmenting, since a is already saturated by M.

Question 3

Question



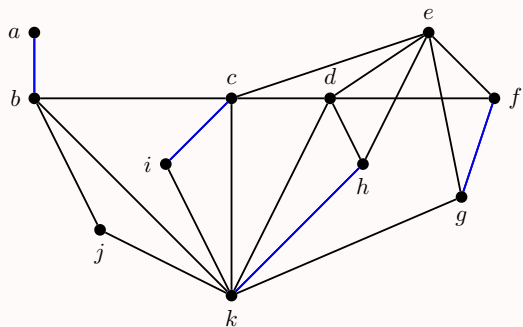
2. Find an augmenting path in the graph or explain why none exists.

Solution.

2. Possible augmenting paths are $jkhe$ and $jkhd$.

Question 3

Question



3. Is M a maximum matching? maximal matching? perfect matching?

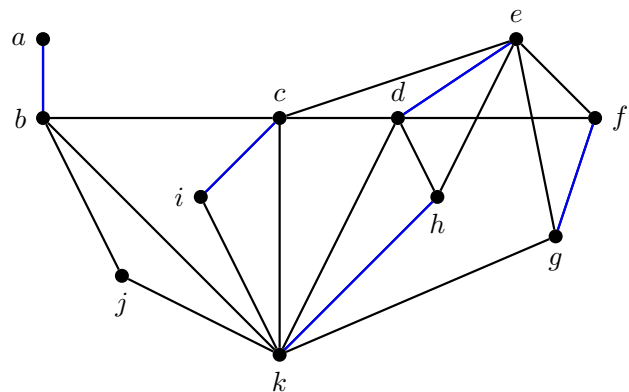
Solution.

- M is not maximum, since an augmenting path exists (Berge's Theorem)
- It is not maximal, because we can enlarge it by adding the edge ed.
- Not perfect because it does not saturate all vertices

Question 3

Solution.

- There are 11 vertices, maximum possible size of a matching is 5

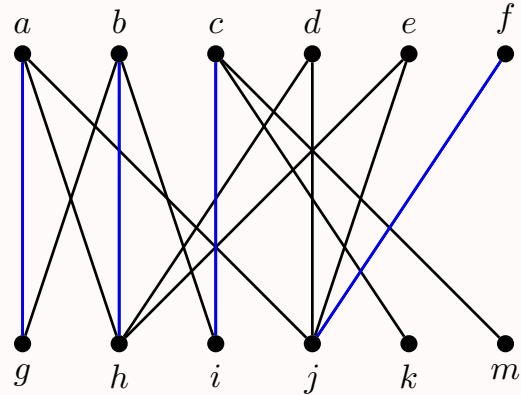


Question 4 - 1

Question

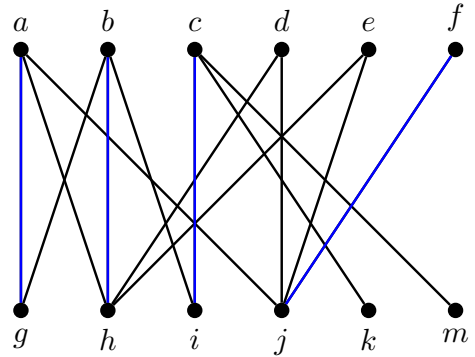
(iii) Use the Augmenting Path Algorithm to find a maximum matching.

(iv) Use the Vertex Cover Method to find a minimum vertex cover.



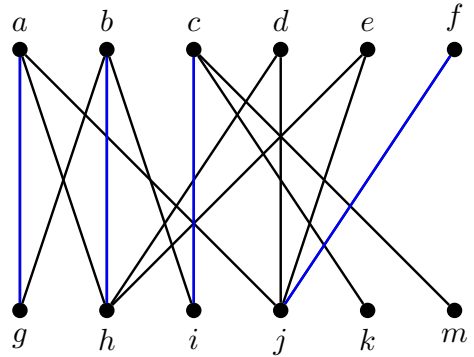
Question 4 - 1

- Step 2. $U = \{d, e\}$
- Step 3. select $d \in U$
- Step 4. neighbors of d: h, j
- Step 6. Possible alternating paths are $dhbg$ and $dhbicm$; the path $dhbicm$ is augmenting.



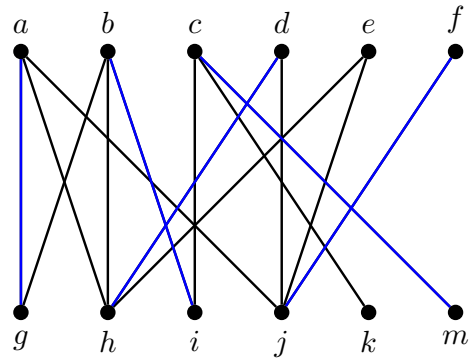
Question 4 - 1

- Step 2. $U = \{d, e\}$
- Step 3. select $d \in U$
- Step 4. neighbors of d: h, j
- Step 6. The augmenting path is $dhbicm$. Switch the edges along this path: remove bh and ci, and add dh, bi, and cm.



Question 4 - 1

- Step 2. $U = \{e\}$
- Step 3. select $e \in U$
- Step 4. neighbors of e: h, j
- Step 6. $ehdjf$, ejf
- Mark vertices: e, h, j, d, f
- Unmarked vertices in X: a, b, c
- Marked vertices in Y: h, j
- Hence a minimum vertex cover is $\{a, b, c, h, j\}$.

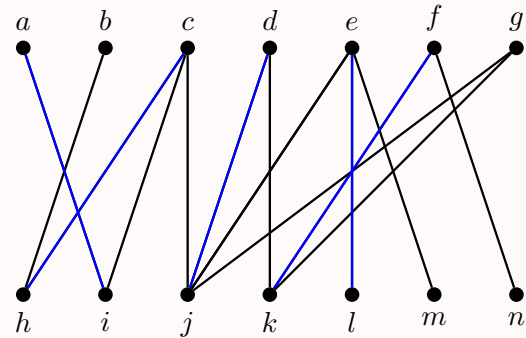


Question 4 - 2

Question

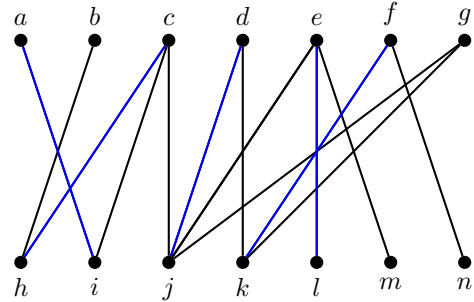
(iii) Use the Augmenting Path Algorithm to find a maximum matching.

(iv) Use the Vertex Cover Method to find a minimum vertex cover.



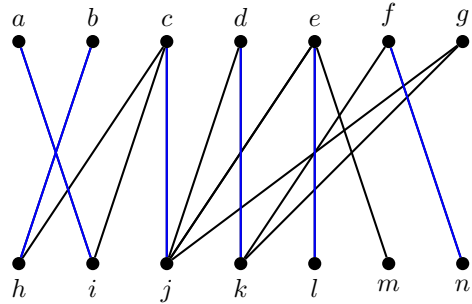
Question 4 - 2

- Step 2. $U = \{b, g\}$
- Step 3. select $b \in U$
- Step 4. neighbor of b: h
- Step 6. $bhcjdkfn$ - augmenting, remove edges ch, jd, kf, add edges bh, cj, dk, fn



Question 4 - 2

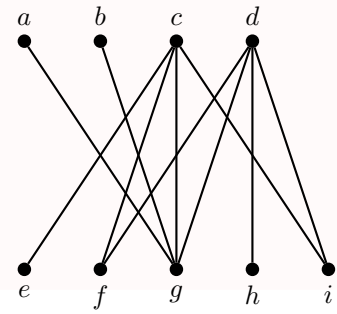
- Step 2. $U = \{g\}$
- Step 3. select $g \in U$
- Step 4. neighbors of g: j, k
- Step 6. $gjchb$, $gjcia$, $gkdjcia$, $gkdjchb$ – not augmenting
- Mark vertices: $g, j, k, c, h, b, i, a, d$
- Unmarked vertices in X: e, f
- Marked vertices in Y: h, i, j, k
- Minimum vertex cover: e, f, h, i, j, k



Questions 5 and 6 – 1

Question

- Find a maximum matching
- Is the graph bipartite?
- If yes, find a minimum vertex cover



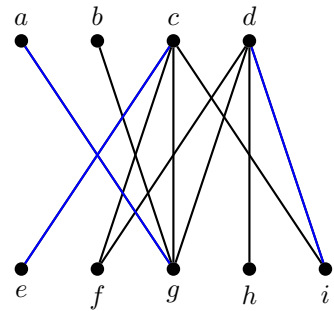
Solution.

- Since both a and b are adjacent only to g, the size of a matching is at most 3.
- A possible maximum matching: ag, ce, di
- The graph is bipartite
- Now, given the maximum matching, we can apply Augmenting Path Algorithm starting from step 2 to find a minimum vertex cover

Questions 5 and 6 – 1

Solution.

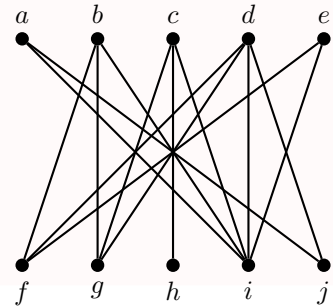
- A possible maximum matching: ag, ce, di
- $U = \{b\}$, neighbor of b is g
- Alternating path: bga
- Marked vertices: b, g, a
- Therefore, a minimum vertex cover is $\{c, d, g\}$.



Questions 5 and 6 – 2

Question

- Find a maximum matching
- Is the graph bipartite?
- If yes, find a minimum vertex cover



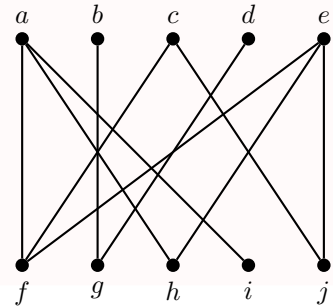
Solution.

- There are five vertices in each part, so the maximum possible size of a matching is 5.
- A possible maximum matching: bf, dg, aj, ch, ei
- The graph is bipartite, and we have a perfect matching
- Vertex cover: a, b, c, d, e

Questions 5 and 6 – 3

Question

- Find a maximum matching
- Is the graph bipartite?
- If yes, find a minimum vertex cover



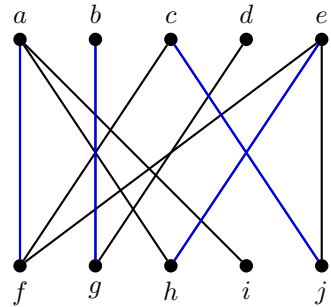
Solution.

- Since both b and d are adjacent only to g, the maximum possible size of a matching is 4.
- A possible maximum matching: bg, af, cj, eh
- The graph is bipartite
- Now, given the maximum matching, we can apply Augmenting Path Algorithm

Questions 5 and 6 – 3

Solution.

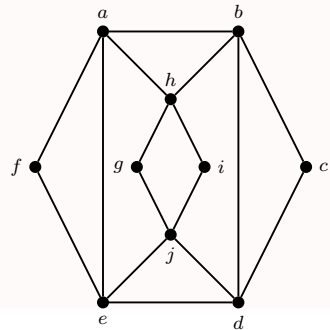
- A possible maximum matching: bg, af, cj, eh
- $U = \{d\}$
- The only alternating path to consider is dgb, and it is not augmenting.
- Vertex cover is a, c, e, g



Questions 5 and 6 – 4

Question

- Find a maximum matching
- Is the graph bipartite?
- If yes, find a minimum vertex cover



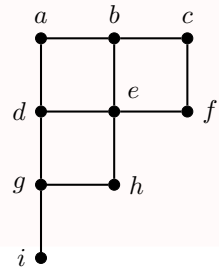
Solution.

- There are in total 10 vertices, maximum possible size of a matching is five
- A maximum matching is $\{ab, fe, cd, hi, gj\}$.
- Not bipartite, odd cycle $afea$

Questions 5 and 6 – 5

Question

- Find a maximum matching
- Is the graph bipartite?
- If yes, find a minimum vertex cover

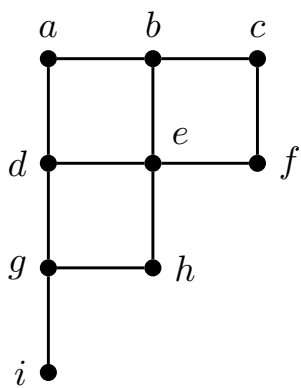


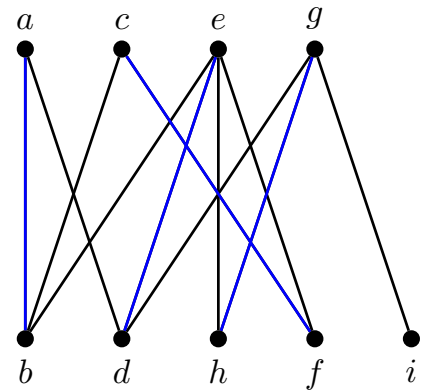
Solution.

- There are 9 vertices in total, so the maximum possible size of a matching is 4.
- ab, cf, ed, gh
- Bipartite

Questions 5 and 6 – 5

Solution.



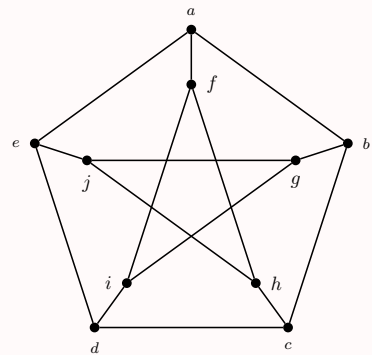


- ab, cf, ed, gh
- Take the lower vertices as X
- $U = \{i\}$. An alternating path is $ighedabcf$. Hence a minimum vertex cover is $\{a, c, e, g\}$.

Questions 5 and 6 – 6

Question

- Find a maximum matching
- Is the graph bipartite?
- If yes, find a minimum vertex cover



Solution.

- There are in total 10 vertices, maximum possible size of a matching is five
- af, bg, hc, di, ej
- Not bipartite, odd cycle $aejgba$

Question 10

Question

Assign two courses to each professor

Professor	Preferred Courses		
Dave	Abstract Algebra	Real Analysis	Number Theory
	Calculus II	Calculus I	Statistics
Roland	Vector Calculus	Discrete Math	Statistics
	Calculus II	Geometry	Calculus I
Chris	Vector Calculus	Real Analysis	Discrete Math
	Statistics	Geometry	Calculus I
Adam	Statistics	Calculus I	Number Theory
	Geometry	Differential Equations	
Hannah	Abstract Algebra	Real Analysis	Number Theory
	Linear Algebra	Topology	
Maggie	Abstract Algebra	Real Analysis	Linear Algebra
	Geometry	Topology	Calculus II

Question 10

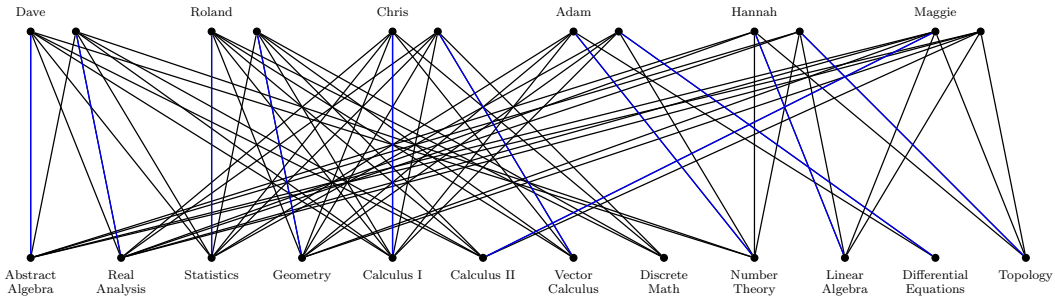
Solution. To model this as a perfect matching problem, we use the following approach:

- Vertices in X represent professor-copies for each professor, create two vertices
- Vertices in Y represent courses
- Draw an edge between $x \in X$ and $y \in Y$ if y is on the preference list of x.

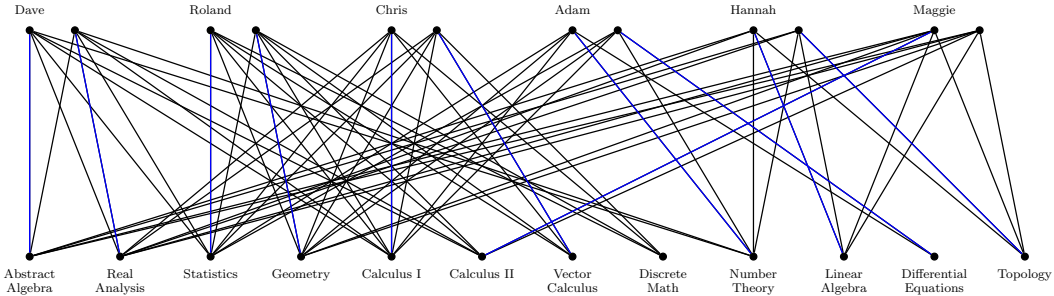
There are six professors, hence 12 vertices in X, and 12 courses, hence 12 vertices in Y. The solution to the problem is a perfect matching for the bipartite graph.

Question 10

- One possible solution: we can start by matching the first vertex available for each course



Question 10



- Discrete Math and one copy of Maggie are unsaturated.
- Roland is currently matched to Geometry but can also teach Discrete Math.
- Replace Roland–Geometry with Roland–Discrete Math, and match the remaining copy of Maggie to Geometry.
- The resulting matching is perfect, since all 12 professor-copies and all 12 courses are saturated.

Tutorial 12

- Matching and vertex cover
- Distinct representatives
- Matchings in Non-Bipartite
- Gale-Shapley Algorithm

Distinct representatives

Definition

Given a collection of finite nonempty sets $S_1, S_2, \ldots, S_n$ (where $n \geq 1$), a system of *distinct representatives* is a collection $r_1, r_2, \ldots, r_n$ such that

$$r_i \in S_i, \quad r_i \neq r_j \text{ if } i \neq j$$

for all $i, j = 1, 2, \ldots, n$.

- In less technical terms, the idea of distinct representatives is that a collection of groups each need their own representative and no two groups can have the same representative

Distinct representatives – example

Example

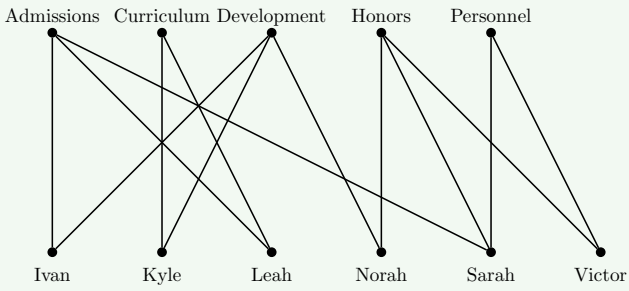
Committee	**Members**		
Admissions Council	Ivan	Leah	Sarah
Curriculum Committee	Kyle	Leah	
Development and Grants	Ivan	Kyle	Norah
Honors Program Council	Norah	Sarah	Victor
Personnel Committee	Sarah	Victor	

- Bipartite graph: X consists of the committees and Y the members
- Edges: a person is a member of that committee
- A collection of distinct representatives is a matching

Distinct representatives – example

Example

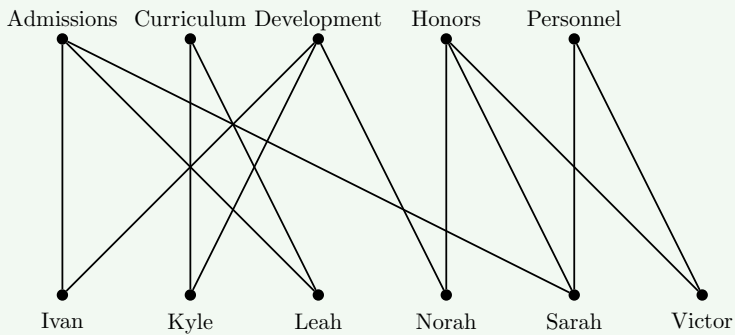
Committee	**Members**		
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Distinct representatives – example

Example

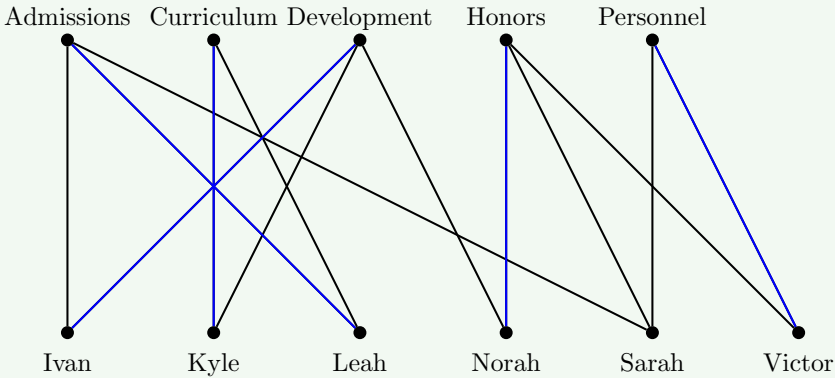
- A collection of distinct representatives is modeled as a matching
- We are interested in each committee having a representative, not every person being a representative.
- Thus we want an X-matching



Distinct representatives – example

Example

- A possible matching



Tutorial 12

- Matching and vertex cover
- Distinct representatives
- Matchings in Non-Bipartite
- Gale-Shapley Algorithm

Canoe example

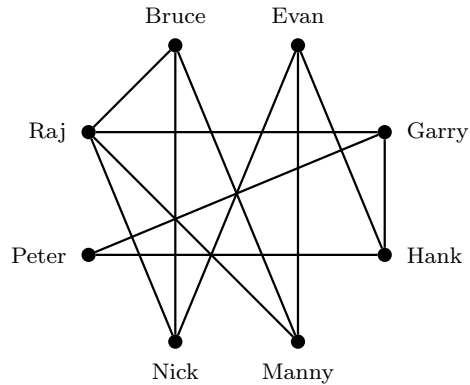
- Eight men, Y indicates a possible pair to share a canoe

	Bruce	Evan	Garry	Hank	Manny	Nick	Peter	Raj
Bruce	.	.	.	.	Y	Y	.	Y
Evan	.	.	.	Y	Y	Y	.	.
Garry	.	.	.	Y	.	.	Y	Y
Hank	.	Y	Y	.	.	.	Y	.
Manny	Y	Y	.	.	.	.	.	Y
Nick	Y	Y	.	.	.	.	.	Y
Peter	.	.	Y	Y	.	.	.	.
Raj	Y	.	.	Y	Y	Y	.	.

- We can model this information as a graph
- Perfect matching: solution to share canoes

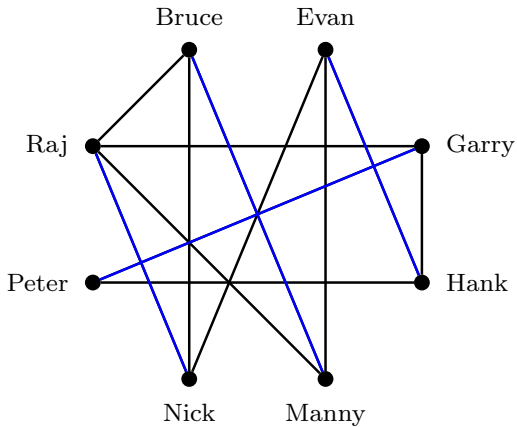
Canoe example

- Peter can only be paired with either Garry or Hank
- Choose to pair Peter and Garry, then Hank must be paired with Evan
- Nick and Manny: each be paired with one of Raj and Bruce



Canoe example

- One possible matching



Matching in general graphs

- Finding matchings in general graphs is often more complex than in bipartite graphs
- Berge' s Theorem holds (M is maximum if and only if G has no M-augmenting paths)
- Augmenting Path Algorithm only applies to bipartite graphs

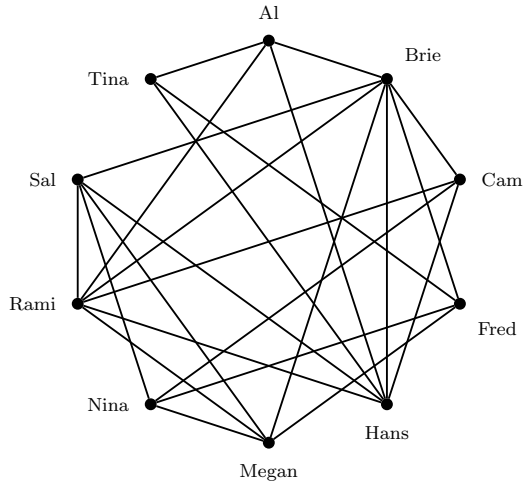
Question 11

Question

	Al	Brie	Cam	Fred	Hans	Megan	Nina	Rami	Sal	Tina
Al	.	Y	.	.	Y	.	.	Y	.	Y
Brie	Y	.	Y	Y	Y	Y	.	Y	Y	.
Cam	.	Y	.	.	Y	.	Y	Y	.	.
Fred	.	Y	.	.	.	Y	Y	.	.	Y
Hans	Y	Y	Y	.	.	.	.	Y	Y	Y
Megan	.	Y	.	Y	.	.	Y	Y	Y	.
Nina	.	.	Y	Y	.	Y	.	.	Y	.
Rami	Y	Y	Y	.	Y	Y	.	.	Y	.
Sal	.	Y	.	.	Y	Y	Y	Y	.	.
Tina	Y	.	.	Y	Y	.	.	.	.	.

Question 11

- We start by matching each vertex to the first vertex on the right
- Tina–Al, Brie–Cam, Fred–Megan, Hans–Rami, Nina–Sal.



Tutorial 12

- Matching and vertex cover
- Distinct representatives
- Matchings in Non-Bipartite
- Gale-Shapley Algorithm

Gale-Shapley Algorithm (with Unacceptable Partners)

- We still model the situation using a bipartite graph, but now the graph need not be complete
- Some people may regard certain partners as unacceptable, so not every pair is allowed
- As a result, a stable matching may leave some people unmatched
- A matching is *stable* if there is no unmatched pair x and y such that
  - x and y are acceptable to each other, and
  - each is either unmatched or prefers the other to their current partner
- Input: preference rankings of n women and n men
- Output: a stable matching

Gale-Shapley Algorithm (with Unacceptable Partners) – steps

1. Each unmatched man proposes to the highest ranked acceptable woman on his list who has not rejected him.
2. If every woman receives at most one acceptable proposal, each such proposal is accepted and the resulting matching is stable. Otherwise go to Step 3.
3. Each woman who receives proposals
   i. rejects any proposal from an unacceptable man;
   ii. keeps her most preferred acceptable proposal as "maybe" and rejects the rest.
4. Any rejected man proposes again to the highest ranked acceptable woman on his list who has not rejected him.
5. Repeat Steps 2–4 until everyone is matched or until no unmatched man has any acceptable woman left to propose to.

Remark

- The algorithm always produces a stable matching.
- Some people may remain unmatched.
- The procedure can also be run with women proposing.

Question 12 – men proposing

Question

Alice:	r	$>$	s	$>$	t	$>$	v	Rich:	a	$>$	d	$>$	b	$>$	c
Beth:	s	$>$	r	$>$	v	$>$	t	Stefan:	a	$>$	c	$>$	d	$>$	b
Cindy:	v	$>$	t	$>$	r	$>$	s	Tom:	c	$>$	b	$>$	d	$>$	a
Dahlia:	t	$>$	v	$>$	s	$>$	r	Victor:	c	$>$	d	$>$	b	$>$	a

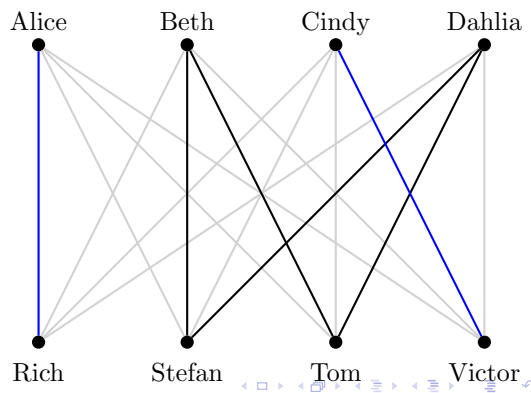
- Step 1. Rich-Alice, Stefan-Alice, Tom-Cindy, Victor-Cindy
- Step 3. Alice accepts Rich and rejects Stefan; Cindy accepts Victor and rejects Tom.

Question 12 – men proposing

Alice:	r	>	s	>	t	>	v	
Beth:	s	>	r	>	v	>	t	
Cindy:	v	>	t	>	r	>	s	
Dahlia:	t	>	v	>	s	>	r	

Rich:	a	>	d	>	b	>	c	
Stefan:	a	>	c	>	d	>	b	
Tom:	c	>	b	>	d	>	a	
Victor:	c	>	d	>	b	>	a	

- Step 4. Stefan - Dahlia, Tom - Beth
- Step 2. both proposals are different, both women accept the proposals
- Stable matching: Rich-Alice, Victor-Cindy, Stefan - Dahlia, Tom - Beth



Question 12 – women proposing

Question

Alice:	r > s > t > v	Rich:	a > d > b > c
Beth:	s > r > v > t	Stefan:	a > c > d > b
Cindy:	v > t > r > s	Tom:	c > b > d > a
Dahlia:	t > v > s > r	Victor:	c > d > b > a

- Step 1. Alice-Rich, Beth-Stefan, Cindy-Victor, Dahlia-Tom
- Step 2. All proposals are different, so all men accept them.

Egalitarian cost

- The *pairwise egalitarian cost* of a man and a woman is the sum of the rankings they give each other
- The *egalitarian cost* of a stable matching is the sum of all the pairwise egalitarian costs of the married couples in the matching
- For comparison of matchings

Question 12 – egalitarian cost

Alice:	r	$>$	s	$>$	t	$>$	v	Rich:	a	$>$	d	$>$	b	$>$	c
Beth:	s	$>$	r	$>$	v	$>$	t	Stefan:	a	$>$	c	$>$	d	$>$	b
Cindy:	v	$>$	t	$>$	r	$>$	s	Tom:	c	$>$	b	$>$	d	$>$	a
Dahlia:	t	$>$	v	$>$	s	$>$	r	Victor:	c	$>$	d	$>$	b	$>$	a

- Men proposing: Rich-Alice, Victor-Cindy, Stefan-Dahlia, Tom-Beth

$$(1+1)+(1+1)+(3+3)+(2+4) = 2+2+6+6 = 16$$

- Women proposing: Alice-Rich, Beth-Stefan, Cindy-Victor, Dahlia-Tom

$$(1+1)+(1+4)+(1+1)+(1+3) = 2+5+2+4 = 13$$

Question 13 – men proposing

Question

Edith:	l > n > o > m > p	Liam:	f > e > h > g > i
Faye:	n > l > m > o > p	Malik:	e > i > g > f > h
Grace:	p > m > o > n > l	Nate:	f > g > i > h > e
Hanna:	p > n > o > l > m	Olaf:	i > e > f > g > h
Iris:	p > o > m > n > l	Pablo:	f > h > g > e > i

- Step 1. Liam-Faye, Malik-Edith, Nate-Faye, Olaf-Iris, Pablo-Faye
- Step 3. Faye accepts Nate and rejects the rest; Edith says "maybe" to Malik; Iris says "maybe" to Olaf.

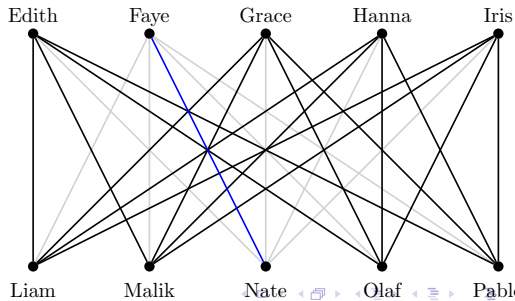
Question 13 – men proposing

Question

Edith:	l	>	n	>	o	>	m	>	p
Faye:	n	>	l	>	m	>	o	>	p
Grace:	p	>	m	>	o	>	n	>	l
Hanna:	p	>	n	>	o	>	l	>	m
Iris:	p	>	o	>	m	>	n	>	l

Liam:	f	>	e	>	h	>	g	>	i
Malik:	e	>	i	>	g	>	f	>	h
Nate:	f	>	g	>	i	>	h	>	e
Olaf:	i	>	e	>	f	>	g	>	h
Pablo:	f	>	h	>	g	>	e	>	i

- Step 4. Liam-Edith, Malik-Edith, Olaf-Iris, Pablo-Hanna
- Step 3. Edith accepts Liam; Iris says "maybe" to Olaf; Hanna accepts Pablo

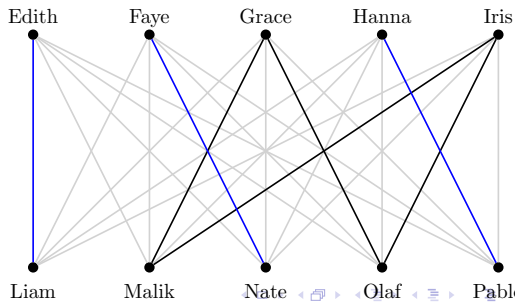


Question 13 – men proposing

Question

Edith:	l > n > o > m > p	Liam:	f > e > h > g > i
Faye:	n > l > m > o > p	Malik:	e > i > g > f > h
Grace:	p > m > o > n > l	Nate:	f > g > i > h > e
Hanna:	p > n > o > l > m	Olaf:	i > e > f > g > h
Iris:	p > o > m > n > l	Pablo:	f > h > g > e > i

- Step 4. Malik-Iris, Olaf-Iris
- Step 3. Iris accepts Olaf and rejects Malik



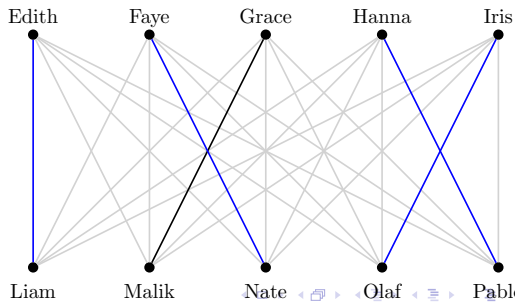
Question 13 – men proposing

Question

Edith:	l	>	n	>	o	>	m	>	p
Faye:	n	>	l	>	m	>	o	>	p
Grace:	p	>	m	>	o	>	n	>	l
Hanna:	p	>	n	>	o	>	l	>	m
Iris:	p	>	o	>	m	>	n	>	l

Liam:	f	>	e	>	h	>	g	>	i
Malik:	e	>	i	>	g	>	f	>	h
Nate:	f	>	g	>	i	>	h	>	e
Olaf:	i	>	e	>	f	>	g	>	h
Pablo:	f	>	h	>	g	>	e	>	i

- Step 4. Malik-Grace
- Step 2. Grace accepts
- Stable matching: Nate-Faye, Liam-Edith, Pablo-Hanna, Olaf-Iris, Malik-Grace



Question 13 – women proposing

Question

Edith:	l > n > o > m > p	Liam:	f > e > h > g > i
Faye:	n > l > m > o > p	Malik:	e > i > g > f > h
Grace:	p > m > o > n > l	Nate:	f > g > i > h > e
Hanna:	p > n > o > l > m	Olaf:	i > e > f > g > h
Iris:	p > o > m > n > l	Pablo:	f > h > g > e > i

- Step 1. Edith-Liam, Faye-Nate, Grace-Pablo, Hanna-Pablo, Iris-Pablo
- Step 3. Nate accepts Faye; Liam says "maybe" to Edith; Pablo says "maybe" to Hanna, rejects Iris and Grace

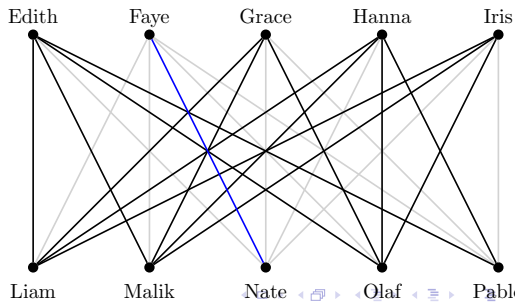
Question 13 – women proposing

Question

Edith:	l	>	n	>	o	>	m	>	p
Faye:	n	>	l	>	m	>	o	>	p
Grace:	p	>	m	>	o	>	n	>	l
Hanna:	p	>	n	>	o	>	l	>	m
Iris:	p	>	o	>	m	>	n	>	l

Liam:	f	>	e	>	h	>	g	>	i
Malik:	e	>	i	>	g	>	f	>	h
Nate:	f	>	g	>	i	>	h	>	e
Olaf:	i	>	e	>	f	>	g	>	h
Pablo:	f	>	h	>	g	>	e	>	i

- Step 4. Edith-Liam, Grace-Malik, Hanna-Pablo, Iris-Olaf
- Step 2. All proposals are different, so all men accept them.
- Stable matching: Faye-Nate, Edith-Liam, Grace-Malik, Hanna-Pablo, Iris-Olaf



Question 13 – egalitarian cost

Edith:	l > n > o > m > p	Liam:	f > e > h > g > i
Faye:	n > l > m > o > p	Malik:	e > i > g > f > h
Grace:	p > m > o > n > l	Nate:	f > g > i > h > e
Hanna:	p > n > o > l > m	Olaf:	i > e > f > g > h
Iris:	p > o > m > n > l	Pablo:	f > h > g > e > i

- Men proposing: Nate-Faye, Liam-Edith, Olaf-Iris, Pablo-Hanna, Malik-Grace

$$(1+1)+(2+1)+(1+2)+(2+1)+(3+2) = 2+3+3+3+5 = 16$$

- Women proposing: Edith-Liam, Faye-Nate, Grace-Malik, Hanna-Pablo, Iris-Olaf

$$(1+2)+(1+1)+(2+3)+(1+2)+(2+1) = 3+2+5+3+3 = 16$$

Gale-Shapley Algorithm (with Unacceptable Partners)

- We are still looking for a matching in a bipartite graph, only now the graph is not complete
- We must adjust our notion of a stable matching, since it is possible that not all people could be matched
- Under these new conditions, a matching (with unacceptable partners) is *stable* if no unmatched pair x and y such that x and y are both acceptable to each other, and each is either single or prefers the other to their current partner
- Input: preference ranking of n women and n men
- Output: stable matching

Gale-Shapley Algorithm (with Unacceptable Partners) – steps

1. Each man proposes to the highest ranking woman on his list
2. If every woman receives only one proposal from someone they deem acceptable, they all accept and this matching is stable. Otherwise move to step 3
3. Each woman
   i. rejects a proposal if it is from an unacceptable man;
   ii. accepts if the proposal is from the man she prefers above all other currently available men and rejects the rest (if any); or
   iii. delays with a maybe to the highest ranked proposal and rejects the rest (if any)
4. Each man now proposes to the highest ranking unmatched woman on their list who has not rejected him
5. Repeat steps 2-4 until all people have been paired or until no unmatched man has any acceptable partners remaining

Remark

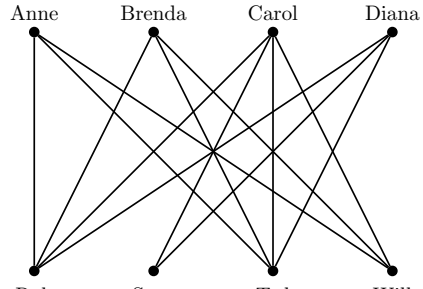
- Always produces a stable matching
- Can be modified so that the women are proposing

Question 14 – women proposing

Question

Anne:	t > r > w	Rob:	a > b > c > d
Brenda:	w > r > t	Stan:	a > b
Carol:	w > r > s > t	Ted:	c > d > a > b
Diana:	s > r > t	Will:	c > b > a

- Step 1. Anne-Ted, Brenda-Will, Carol-Will, Diana-Stan
- Step 3. Ted says "maybe" to Anne; Will accepts Carol and rejects Brenda; Stan rejects Diana

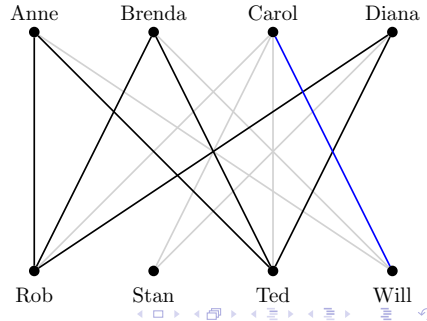


Question 14 – women proposing

Question

Anne:	t > r > w	Rob:	a > b > c > d
Brenda:	w > r > t	Stan:	a > b
Carol:	w > r > s > t	Ted:	c > d > a > b
Diana:	s > r > t	Will:	c > b > a

- Step 4. Anne-Ted, Brenda-Rob, Diana-Rob
- Step 3. Ted says "maybe" to Anne; Rob says "maybe" to Brenda and rejects Diana

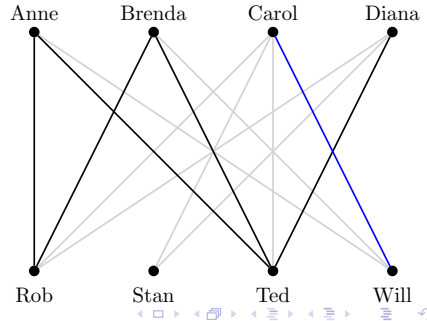


Question 14 – women proposing

Question

Anne:	t > r > w	Rob:	a > b > c > d
Brenda:	w > r > t	Stan:	a > b
Carol:	w > r > s > t	Ted:	c > d > a > b
Diana:	s > r > t	Will:	c > b > a

- Step 4. Anne-Ted, Brenda-Rob, Diana-Ted
- Step 3. Ted accepts Diana and rejects Anne; Rob says "maybe" to Brenda

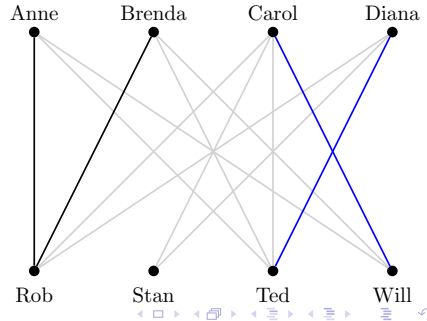


Question 14 – women proposing

Question

Anne:	t > r > w	Rob:	a > b > c > d
Brenda:	w > r > t	Stan:	a > b
Carol:	w > r > s > t	Ted:	c > d > a > b
Diana:	s > r > t	Will:	c > b > a

- Step 4. Anne-Rob, Brenda-Rob
- Step 3. Rob accepts Anne and rejects Brenda
- The resulting stable matching is Carol–Will, Anne–Rob, and Diana–Ted.

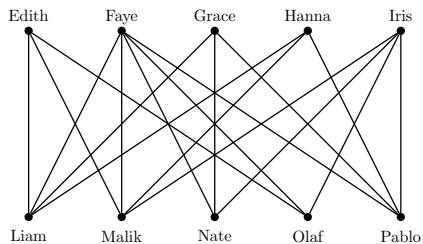


Question 15 – men proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 1. Liam-Faye, Malik-Edith, Nate-Grace, Olaf-Iris, Pablo-Faye
- Step 3. Faye says "maybe" to Liam and rejects Pablo; Edith says "maybe" to Malik; Grace says "maybe" to Nate; Iris rejects Olaf

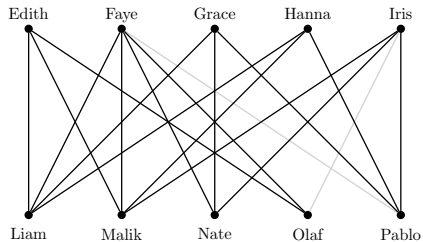


Question 15 – men proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Liam-Faye, Malik-Edith, Nate-Grace, Olaf-Edith, Pablo-Hanna
- Step 3. Faye says "maybe" to Liam; Edith says "maybe" to Malik and rejects Olaf; Grace says "maybe" to Nate; Hanna accepts Pablo

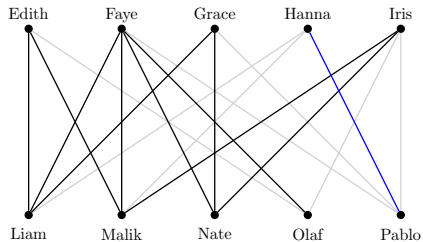


Question 15 – men proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Liam-Faye, Malik-Edith, Nate-Grace, Olaf-Faye
- Step 3. Faye says "maybe" to Liam and rejects Olaf; Edith says "maybe" to Malik; Grace says "maybe" to Nate

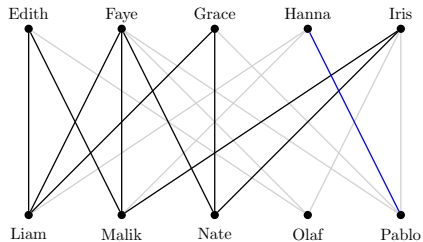


Question 15 – men proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Liam-Faye, Malik-Edith, Nate-Grace
- Step 2. Every woman accepts the proposal.
- Stable matching: Pablo-Hanna, Liam-Faye, Malik-Edith, Nate-Grace

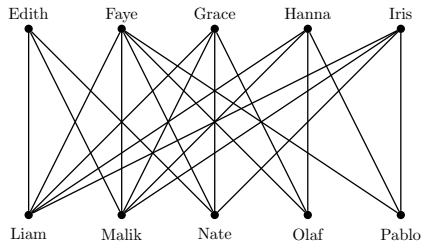


Question 15 – women proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 1. Edith-Liam, Faye-Nate, Grace-Malik, Hanna-Pablo, Iris-Pablo
- Step 3. Liam says "maybe" to Edith; Nate says "maybe" to Faye; Malik rejects Grace; Pablo says "maybe" to Hanna and rejects Iris

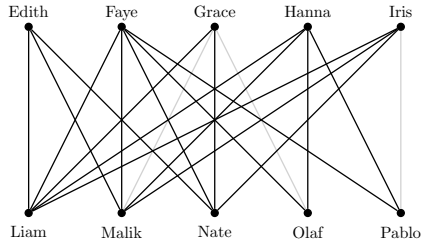


Question 15 – women proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Edith-Liam, Faye-Nate, Grace-Olaf, Hanna-Pablo, Iris-Malik
- Step 3. Liam says "maybe" to Edith; Nate says "maybe" to Faye; Olaf rejects Grace; Pablo says "maybe" to Hanna; Malik says "maybe" to Iris

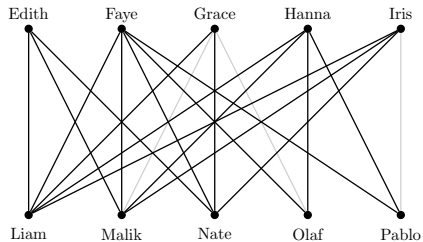


Question 15 – women proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Edith-Liam, Faye-Nate, Grace-Nate, Hanna-Pablo, Iris-Malik
- Step 3. Liam says "maybe" to Edith; Nate accepts Grace and rejects Faye; Pablo says "maybe" to Hanna; Malik says "maybe" to Iris

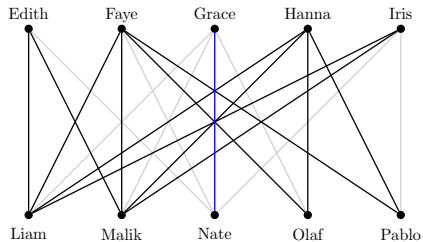


Question 15 – women proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Edith-Liam, Faye-Liam, Hanna-Pablo, Iris-Malik
- Step 3. Liam accepts Faye and rejects Edith; Pablo says "maybe" to Hanna; Malik says "maybe" to Iris

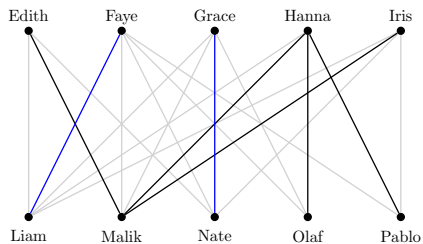


Question 15 – women proposing

Question

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Step 4. Edith-Malik, Hanna-Pablo, Iris-Malik
- Step 3. Malik accepts Edith and rejects Iris; Pablo accepts Hanna
- Stable matching: Faye-Liam, Grace-Nate, Edith-Malik, Hanna-Pablo



Question 15 – egalitarian cost

Edith:	l > n > m	Liam:	f > e > h > g
Faye:	n > l > m > o > p	Malik:	e > h > i > f
Grace:	m > o > n > l	Nate:	g > f > i
Hanna:	p > o > l > m	Olaf:	i > e > f
Iris:	p > m > n > l	Pablo:	f > h > g > i

- Men proposing: Pablo-Hanna, Liam-Faye, Malik-Edith, Nate-Grace

$$(2+1)+(1+2)+(1+3)+(1+3)=3+3+4+4=14$$

- Women proposing: Faye-Liam, Grace-Nate, Edith-Malik, Hanna-Pablo

$$(2+1)+(3+1)+(3+1)+(1+2)=3+4+4+3=14$$

Question 16 – men proposing

Question

Alice:	p > r > s > t	Peter:	b > a > c > d > e
Beth:	r > p > s > t	Rich:	c > b > e > d > a
Carol:	t > p > s > r	Saul:	a > b > c > d > e
Diana:	t > s > r > p	Teddy:	e > c > d > a > b
Edith:	r > s > t > p		

- Since there are five women and four men, we allow one woman to remain unmatched.
- Step 1. Peter-Beth, Rich-Carol, Saul-Alice, Teddy-Edith.
- Step 2. All proposals are different and acceptable, so we obtain a stable matching; Diana remains unmatched.

Question 16 – women proposing

Question

Alice:	p > r > s > t	Peter:	b > a > c > d > e
Beth:	r > p > s > t	Rich:	c > b > e > d > a
Carol:	t > p > s > r	Saul:	a > b > c > d > e
Diana:	t > s > r > p	Teddy:	e > c > d > a > b
Edith:	r > s > t > p		

- Step 1. Alice-Peter, Beth-Rich, Carol-Teddy, Diana-Teddy, Edith-Rich
- Step 3. Peter says "maybe" to Alice; Rich says "maybe" to Beth and rejects Edith; Teddy says "maybe" to Carol and rejects Diana

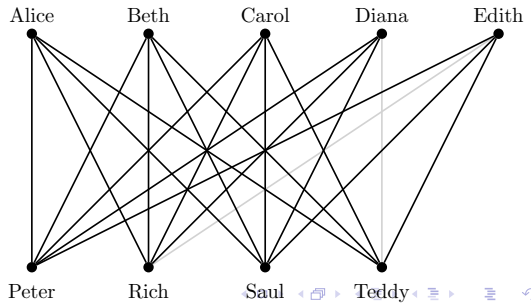
Question 16 – women proposing

Question

Alice:	p	$>$	r	$>$	s	$>$	t		
Beth:	r	$>$	p	$>$	s	$>$	t		
Carol:	t	$>$	p	$>$	s	$>$	r		
Diana:	t	$>$	s	$>$	r	$>$	p		
Edith:	r	$>$	s	$>$	t	$>$	p		

Peter:	b	$>$	a	$>$	c	$>$	d	$>$	e
Rich:	c	$>$	b	$>$	e	$>$	d	$>$	a
Saul:	a	$>$	b	$>$	c	$>$	d	$>$	e
Teddy:	e	$>$	c	$>$	d	$>$	a	$>$	b

- Step 4. Alice-Peter, Beth-Rich, Carol-Teddy, Diana-Saul, Edith-Saul
- Step 3. Peter says “maybe” to Alice; Rich says “maybe” to Beth; Teddy says “maybe” to Carol; Saul says “maybe” to Diana and rejects Edith



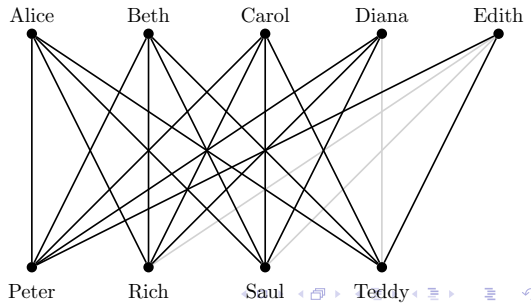
Question 16 – women proposing

Question

Alice:	p	>	r	>	s	>	t		
Beth:	r	>	p	>	s	>	t		
Carol:	t	>	p	>	s	>	r		
Diana:	t	>	s	>	r	>	p		
Edith:	r	>	s	>	t	>	p		

Peter:	b	>	a	>	c	>	d	>	e
Rich:	c	>	b	>	e	>	d	>	a
Saul:	a	>	b	>	c	>	d	>	e
Teddy:	e	>	c	>	d	>	a	>	b

- Step 4. Alice-Peter, Beth-Rich, Carol-Teddy, Diana-Saul, Edith-Teddy
- Step 3. Peter says "maybe" to Alice; Rich says "maybe" to Beth; Teddy accepts Edith and rejects Carol; Saul says "maybe" to Diana



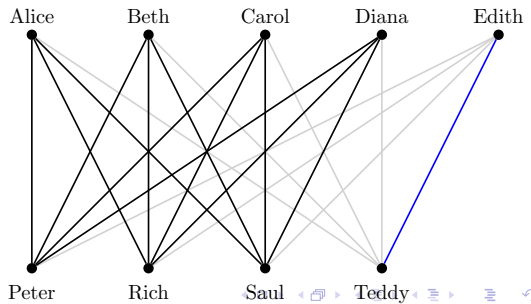
Question 16 – women proposing

Question

Alice:	p	>	r	>	s	>	t		
Beth:	r	>	p	>	s	>	t		
Carol:	t	>	p	>	s	>	r		
Diana:	t	>	s	>	r	>	p		
Edith:	r	>	s	>	t	>	p		

Peter:	b	>	a	>	c	>	d	>	e
Rich:	c	>	b	>	e	>	d	>	a
Saul:	a	>	b	>	c	>	d	>	e
Teddy:	e	>	c	>	d	>	a	>	b

- Step 4. Alice-Peter, Beth-Rich, Carol-Peter, Diana-Saul
- Step 3. Peter says "maybe" to Alice and rejects Carol; Rich says "maybe" to Beth; Saul says "maybe" to Diana



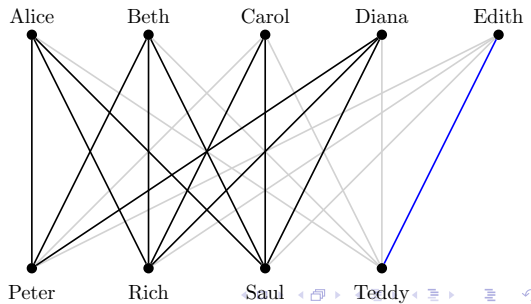
Question 16 – women proposing

Question

Alice:	p	$>$	r	$>$	s	$>$	t		
Beth:	r	$>$	p	$>$	s	$>$	t		
Carol:	t	$>$	p	$>$	s	$>$	r		
Diana:	t	$>$	s	$>$	r	$>$	p		
Edith:	r	$>$	s	$>$	t	$>$	p		

Peter:	b	$>$	a	$>$	c	$>$	d	$>$	e
Rich:	c	$>$	b	$>$	e	$>$	d	$>$	a
Saul:	a	$>$	b	$>$	c	$>$	d	$>$	e
Teddy:	e	$>$	c	$>$	d	$>$	a	$>$	b

- Step 4. Alice-Peter, Beth-Rich, Carol-Saul, Diana-Saul
- Step 3. Peter says "maybe" to Alice; Rich says "maybe" to Beth; Saul says "maybe" to Carol and rejects Diana



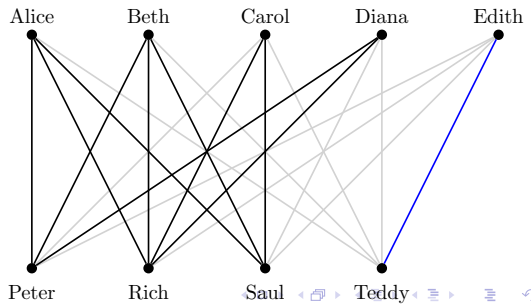
Question 16 – women proposing

Question

Alice:	p	>	r	>	s	>	t			
Beth:	r	>	p	>	s	>	t			
Carol:	t	>	p	>	s	>	r			
Diana:	t	>	s	>	r	>	p			
Edith:	r	>	s	>	t	>	p			

Peter:	b	>	a	>	c	>	d	>	e	
Rich:	c	>	b	>	e	>	d	>	a	
Saul:	a	>	b	>	c	>	d	>	e	
Teddy:	e	>	c	>	d	>	a	>	b	

- Step 4. Alice-Peter, Beth-Rich, Carol-Saul, Diana-Rich
- Step 3. Peter says “maybe” to Alice; Rich says “maybe” to Beth and rejects Diana; Saul says “maybe” to Carol



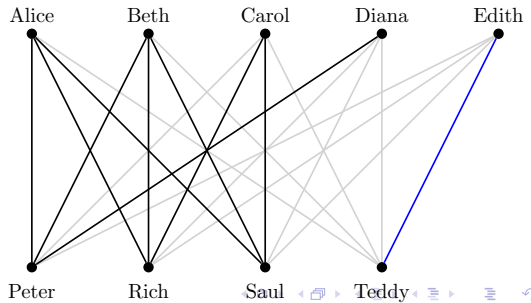
Question 16 – women proposing

Question

Alice:	p	>	r	>	s	>	t		
Beth:	r	>	p	>	s	>	t		
Carol:	t	>	p	>	s	>	r		
Diana:	t	>	s	>	r	>	p		
Edith:	r	>	s	>	t	>	p		

Peter:	b	>	a	>	c	>	d	>	e
Rich:	c	>	b	>	e	>	d	>	a
Saul:	a	>	b	>	c	>	d	>	e
Teddy:	e	>	c	>	d	>	a	>	b

- Step 4. Alice-Peter, Beth-Rich, Carol-Saul, Diana-Peter
- Step 3. Peter says "maybe" to Alice and rejects Diana; Rich says "maybe" to Beth; Saul says "maybe" to Carol



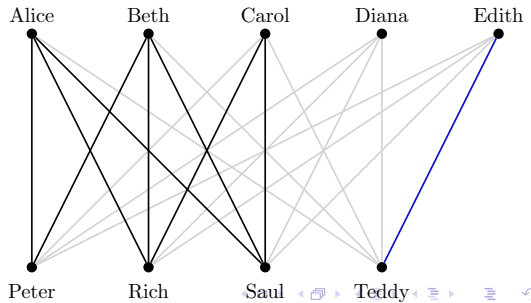
Question 16 – women proposing

Question

Alice:	p	$>$	r	$>$	s	$>$	t		
Beth:	r	$>$	p	$>$	s	$>$	t		
Carol:	t	$>$	p	$>$	s	$>$	r		
Diana:	t	$>$	s	$>$	r	$>$	p		
Edith:	r	$>$	s	$>$	t	$>$	p		

Peter:	b	$>$	a	$>$	c	$>$	d	$>$	e
Rich:	c	$>$	b	$>$	e	$>$	d	$>$	a
Saul:	a	$>$	b	$>$	c	$>$	d	$>$	e
Teddy:	e	$>$	c	$>$	d	$>$	a	$>$	b

- Step 4. Alice-Peter, Beth-Rich, Carol-Saul
- Step 2. All proposals are different
- Stable matching: Edith-Teddy, Alice-Peter, Beth-Rich, Carol-Saul



Question 16 – egalitarian cost

Alice:	p	$>$	r	$>$	s	$>$	t	Peter:	b	$>$	a	$>$	c	$>$	d	$>$	e
Beth:	r	$>$	p	$>$	s	$>$	t	Rich:	c	$>$	b	$>$	e	$>$	d	$>$	a
Carol:	t	$>$	p	$>$	s	$>$	r	Saul:	a	$>$	b	$>$	c	$>$	d	$>$	e
Diana:	t	$>$	s	$>$	r	$>$	p	Teddy:	e	$>$	c	$>$	d	$>$	a	$>$	b
Edith:	r	$>$	s	$>$	t	$>$	p										

- Men proposing: Peter-Beth, Rich-Carol, Saul-Alice, Teddy-Edith

$$(1+2)+(1+4)+(1+3)+(1+3) = 3+5+4+4 = 16$$

- Women proposing: Edith-Teddy, Alice-Peter, Beth-Rich, Carol-Saul

$$(3+1)+(1+2)+(1+2)+(3+3) = 4+3+3+6 = 16$$

Final exam

- **Time**: 16.12.2025, 10:30–13:00
- **Location**: Room -1.40
  - Campus map: https://studuj.fiit.sk/mapa-fakulty/
- **Calculators:** Calculators are permitted, but phone calculators are not allowed.
- Toilet breaks are **not** allowed.
- Write your answers on the provided answer sheets. Additional sheets will be supplied upon request.
- Please write your full name clearly on every page.
- Include detailed computation steps for all solutions. Answers without supporting work will receive zero marks.
- Bring pens in at least two different colors.

Final exam

- Matrix Operators (T6)
- Fundamental Spaces and Decompositions (T7)
- Eulerization and Hierholzer's Algorithm (T8)
- Undirecting Algorithm; find a Hamiltonian cycle using the Nearest Neighbor Algorithm and convert it to a directed Hamiltonian cycle (T9)
- Dijkstra's Algorithm (T10)
- Augmenting Flow Algorithm and Min-Cut Method (T11)

Second final exam

- **Time**: 27.01.2026, 10:30–13:00
- **Location**: Room -1.40
- If you choose to take the second exam, the score from that exam will serve as your final grade.

Second final exam

- Matrix Operators (T6)
- Fundamental Spaces and Decompositions (T7)
- Critical Path and Project Scheduling (T10)
- Minimum Spanning Tree (T10)
  (a) Kruskal's Algorithm
  (b) Prim's Algorithm
- Breadth-First Search Tree and Depth-First Search Tree (T11)
- Gale–Shapley Algorithm to find a stable matching from a set of preferences (T12)